

Transitioning to pangenomics to uncover diversity in Midas Cichlid

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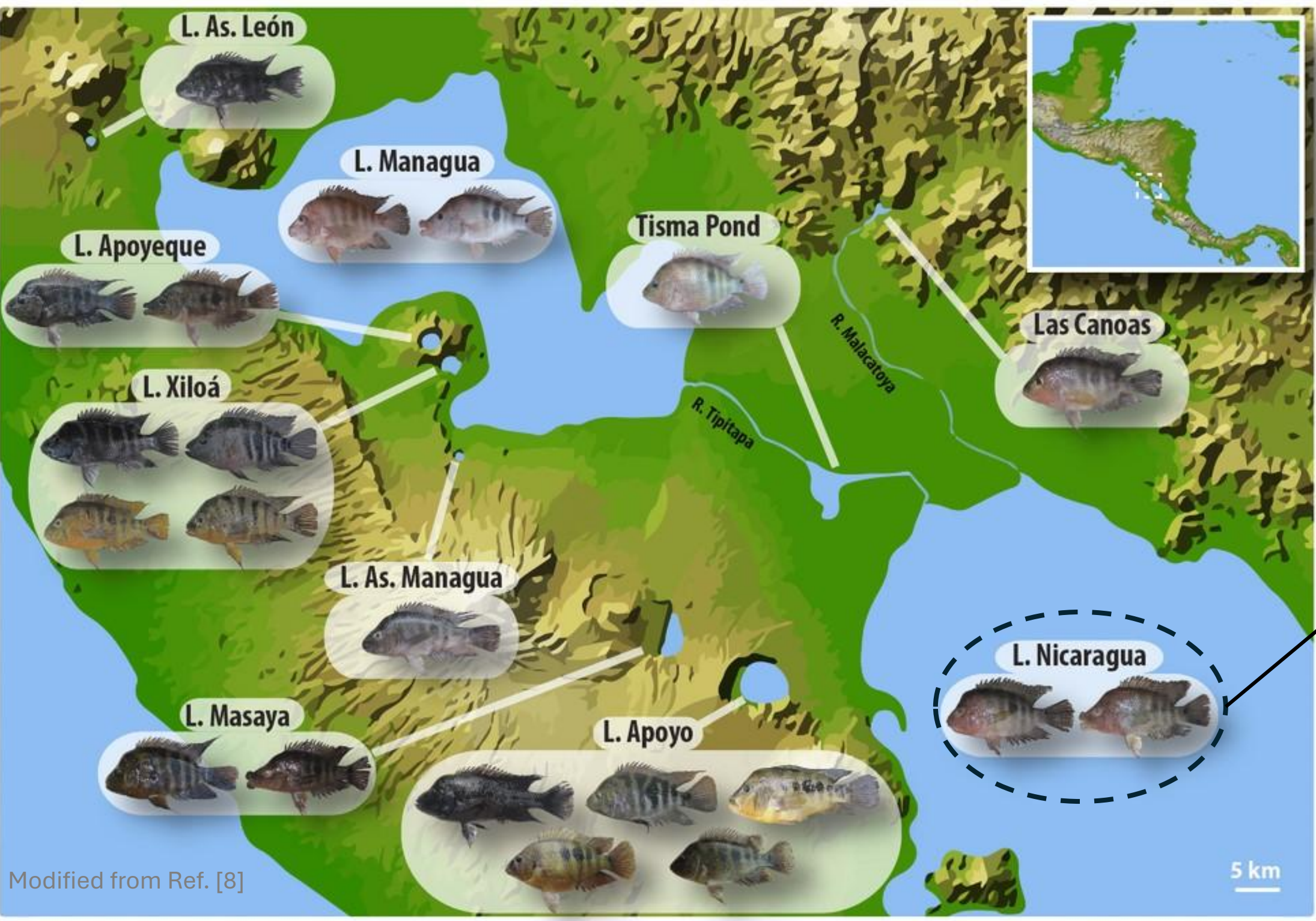
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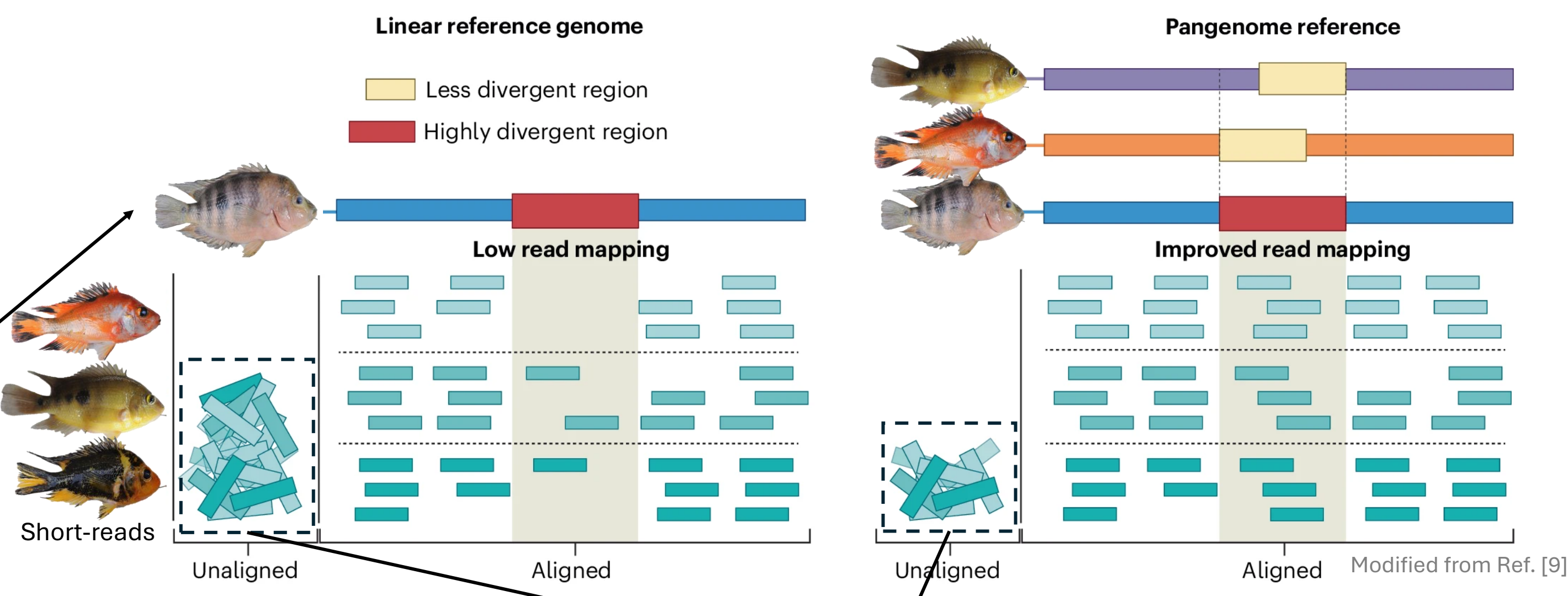
Background

The Neotropical Midas cichlid (*Amphilophus* spp.) is a premier model system for the study of adaptive radiation as well as parallel and sympatric speciation [1]. Our group investigates the evolutionary biology of this species complex through an integrative approach, focusing on eco-morphology [2], genetics [3,4], and host-parasite interactions [5].



Problem: Reference bias

Currently, only one chromosome-level assembly exists, derived from a single individual from Lake Nicaragua, and it lacks functional annotation. To gain a better understanding of this radiation, it is essential to expand the available genomic resources beyond a single reference genome.

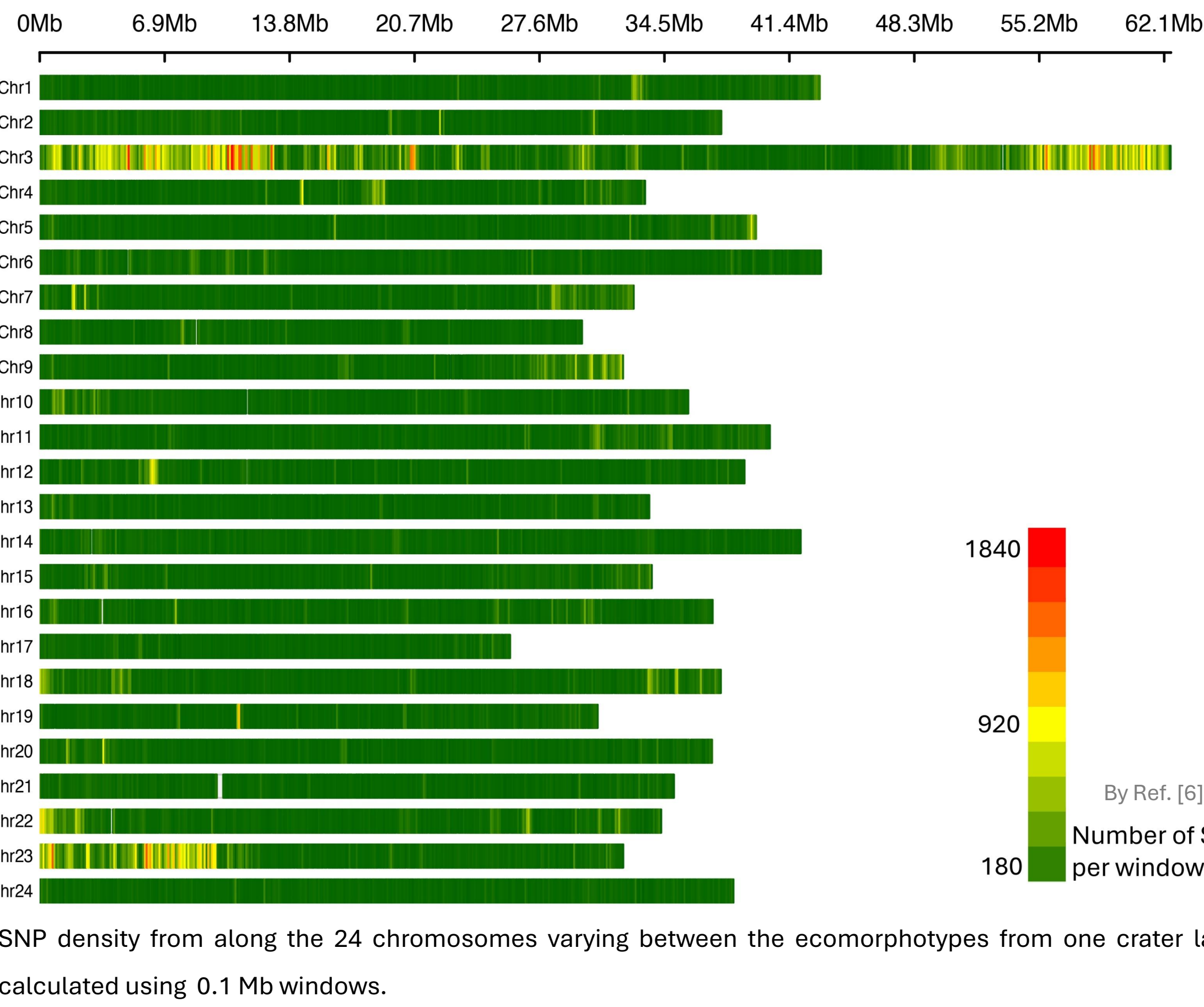


Unaligned sequences from different populations represent **hidden biological diversity**, including Presence-Absence variation (PAV) and Structural Variants (SV) that are often linked to adaptive traits.

Current situation

Population genomics: We performed a variant calling using low-coverage WGS data from multiple populations against the current reference genome [6]. Most of the genomic differentiation is concentrated in specific regions with many immune-related genes.

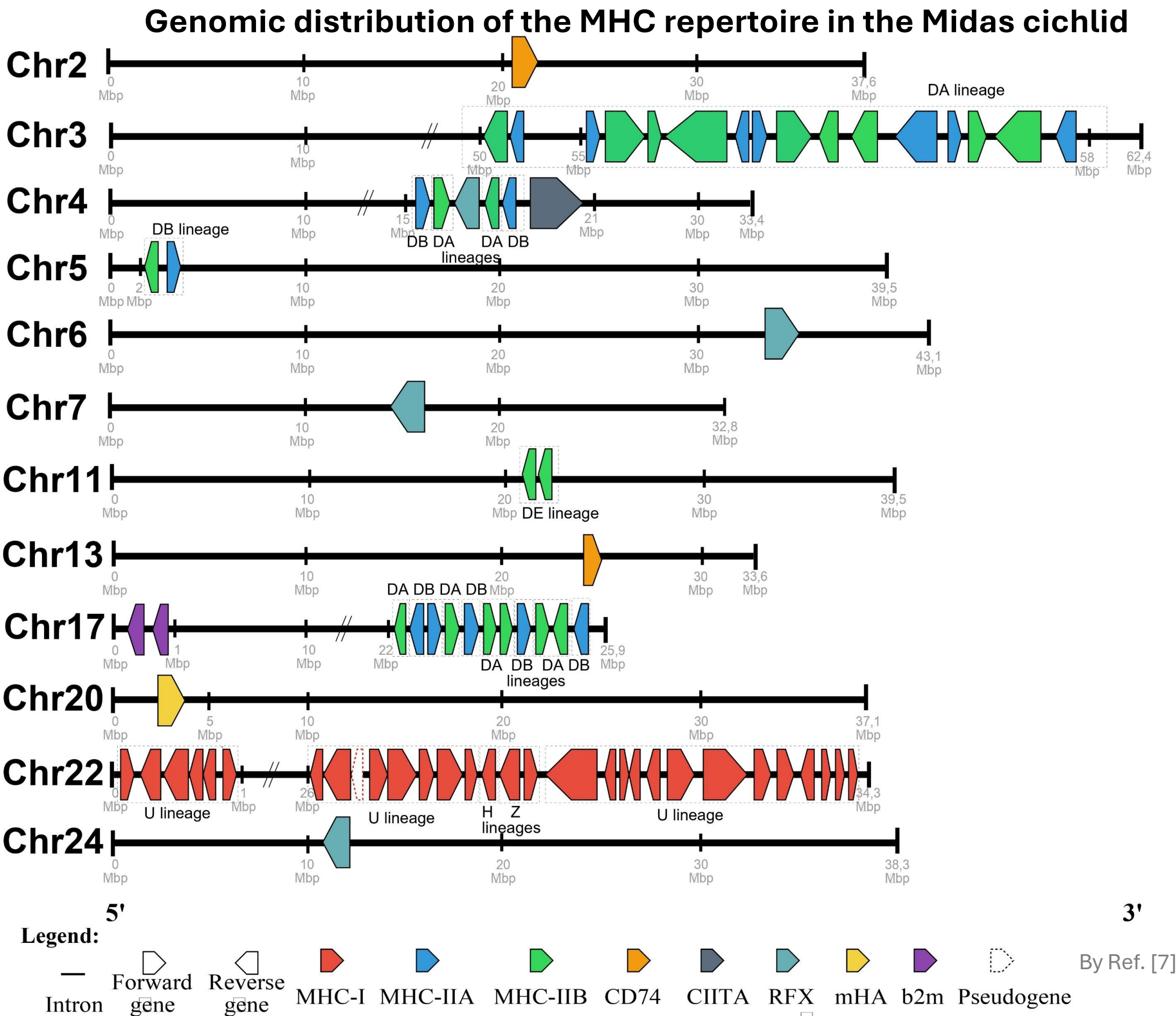
Functional annotation: We developed a custom annotation of the reference genome to identify immune-related genes (i.e. MHC) to better understand host-parasite interactions [7].



Are we losing genetic variance?

Yes. Traditional linear mapping underestimates the true genomic complexity of these diverging populations.

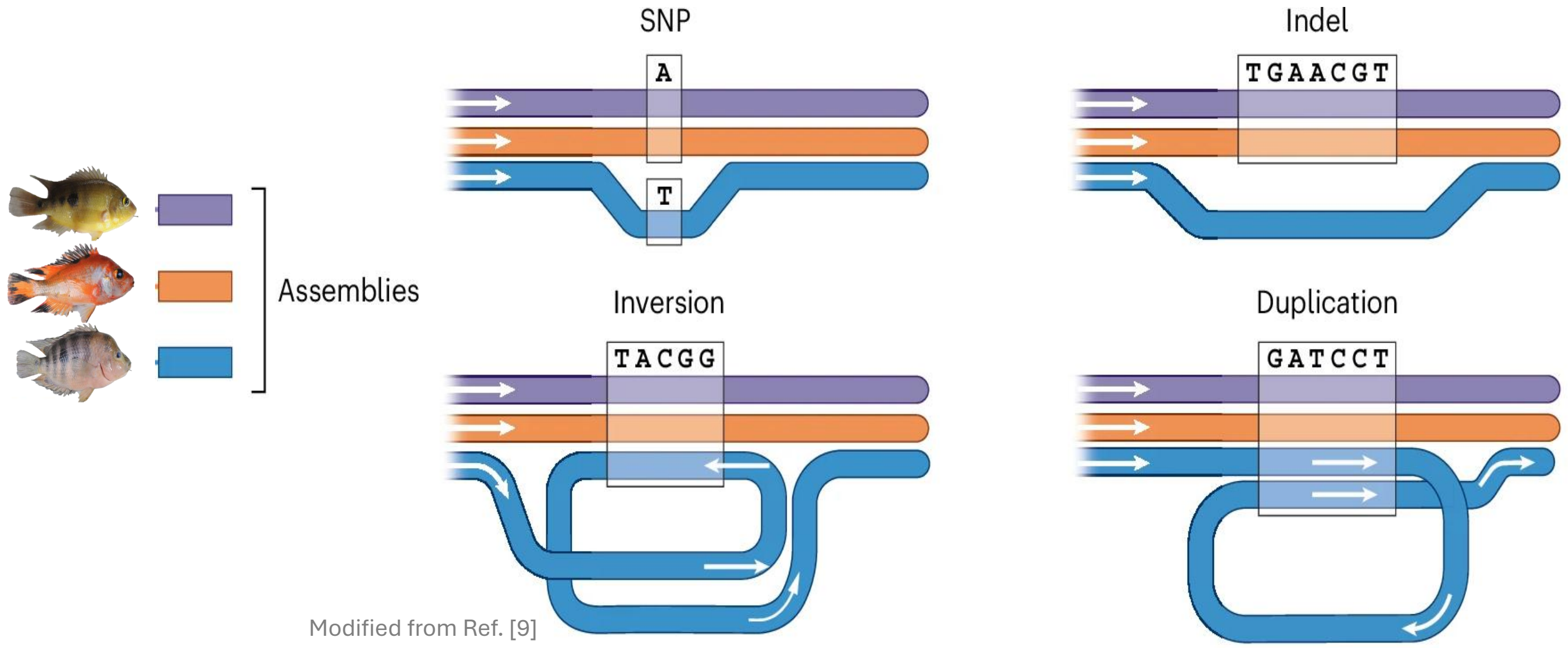
High polymorphism and structural complexity of MHC regions make them ideal targets for pangenomic graphs.



Future directions

Pangenomes integrate the full range of genomic variation within a species complex, overcoming the inherent constraints of single-reference approaches [9,10].

Our objective: Implement a dual-sequencing strategy (PacBio HiFi + Hi-C) to generate de novo chromosome-level assemblies of individuals from different lakes. By constructing a multiassembly graph using the *pggg* ecosystem, we will be able to uncover structural variants within graph bubbles and assess the impact of transposable elements on the genomic plasticity and diversification of the Midas cichlid.



References

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